

Sequences and series - Set 1 - Solutions

Specification MM2.1, MM2.2, MM2.3, M4.19.

Answer key

Question	Answer	Topic
1	B	Sequences and series
2	C	Sequences and series
3	D	Sequences and series
4	A	Sequences and series
5	F	Sequences and series
6	F	Sequences and series
7	B	Sequences and series
8	F	Sequences and series
9	F	Sequences and series
10	C	Sequences and series
11	B	Sequences and series
12	F	Sequences and series

Worked solutions

Question 1 SEQ-01 · Sequences and series · spec MM2.2

An arithmetic series has first term 5 and common difference 3. Find the sum of the first 40 terms.

Answer: B

$$S_{40} = \frac{40}{2}(2 \cdot 5 + 39 \cdot 3) = 20(10 + 117) = 2540.$$

Question 2 SEQ-02 · Sequences and series · spec MM2.3

Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$

Answer: C

$$a = 12, r = -\frac{1}{3}. S_{\infty} = \frac{12}{1 + \frac{1}{3}} = \frac{12}{4/3} = 9.$$

Question 3 SEQ-03 · Sequences and series · spec MM2.1

A sequence is defined by $x_{n+1} = 3x_n - 2$ with $x_1 = 2$. Find x_5 .

Answer: D

$$x_2 = 4, x_3 = 10, x_4 = 28, x_5 = 82.$$

Question 4 SEQ-04 · Sequences and series · spec MM2.2

The n th term of an arithmetic sequence is $u_n = 7 - 4n$. Find the sum of the first 30 terms.

Answer: A

$$u_1 = 3, u_{30} = 7 - 120 = -113. S_{30} = \frac{30}{2}(3 - 113) = 15(-110) = -1650.$$

Question 5 SEQ-06 · Sequences and series · spec MM2.2

The sum of the first n terms of a series is $S_n = 2n^2 + 3n$. Find the 10th term.

Answer: F

$$u_{10} = S_{10} - S_9 = (200 + 30) - (162 + 27) = 230 - 189 = 41.$$

Question 6 SEQ-08 · Sequences and series · spec MM2.3

The second term of a geometric sequence is 6 and the fifth term is 48. Find the sum of the first 6 terms.

Answer: F

$$ar = 6, ar^4 = 48, \text{ so } r^3 = 8, r = 2, a = 3. S_6 = 3 \cdot \frac{2^6 - 1}{2 - 1} = 3 \times 63 = 189.$$

Question 7 SEQ-09 · Sequences and series · spec MM2.2

How many terms of the arithmetic series $3 + 7 + 11 + \dots$ are needed for the sum to first exceed 500?

Answer: B

$$S_n = \frac{n}{2}(6 + 4(n - 1)) = n(2n + 1). n = 15: 465; n = 16: 528 > 500. \text{ So } 16.$$

Question 8 SEQ-10 · Sequences and series · spec MM2.3

A geometric series has first term 2 and common ratio $r > 0$. The sum of the first three terms is 14. Find the sum of the first five terms.

Answer: F

$$2 + 2r + 2r^2 = 14 \Rightarrow r^2 + r - 6 = 0 \Rightarrow r = 2. S_5 = 2(2^5 - 1) = 62.$$

Question 9 SEQ-12 · Sequences and series · spec MM2.3

Which of the following geometric series is/are convergent?

I $1 + 0.9 + 0.81 + \dots$

II $3 - 3 + 3 - 3 + \dots$

III $100 + 10 + 1 + 0.1 + \dots$

Answer: F

A geometric series converges if and only if $|r| < 1$. I has $r = 0.9$, III has $r = 0.1$: both converge. II has $r = -1$ and does not converge.

Question 10 SEQ-13 · Sequences and series · spec MM2.2

An arithmetic sequence has $u_3 = 10$ and $u_8 = 25$. Find the smallest value of n for which $u_n > 100$.

Answer: C

$$5d = 15 \Rightarrow d = 3, a = 4. u_n = 4 + 3(n - 1) = 3n + 1 > 100 \Rightarrow n > 33, \text{ so } n = 34.$$

Question 11 SEQ-14 · Sequences and series · spec MM2.1

The sequence $u_n = n^2 - 10n + 30$ is defined for positive integers n . What is the smallest value taken by any term of the sequence?

Answer: B

$$u_n = (n - 5)^2 + 5, \text{ minimised at } n = 5 \text{ with value } 5.$$

Question 12 SEQ-15 · Sequences and series · spec MM2.3

Evaluate $\sum_{k=1}^{10} 3 \times 2^k$.

Answer: F

This is $6 + 12 + \dots + 6 \cdot 2^9$: first term 6, ratio 2, 10 terms: $6(2^{10} - 1) = 6 \times 1023 = 6138$.